

Financial Institutions and System

Week 11: The Foreign Exchange Market. The International Financial System.

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Agenda

1. The Foreign Exchange Market
2. The International Financial System
3. In-Class Activity and Roadmap



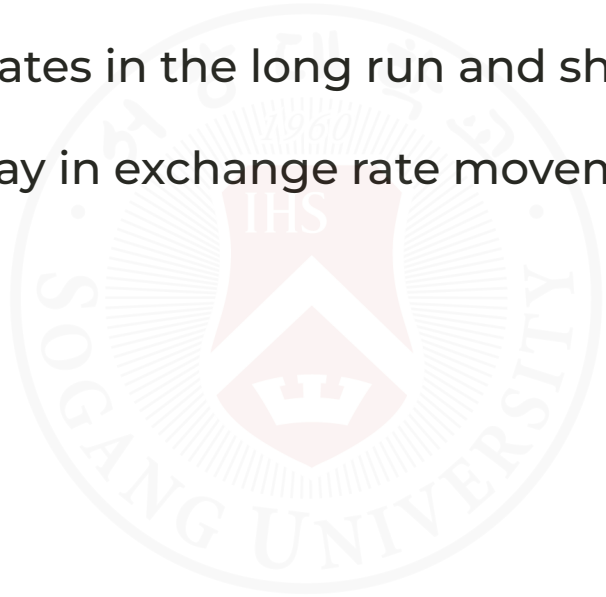
The background of the slide features a large, faint watermark of the Seoul National University (SNU) logo. The logo is circular, with the university's name in Korean and English around the perimeter. In the center, there is a shield with the letters 'IHS' and a crown below it. The year '1960' is also visible.

1. The Foreign Exchange Market

Introduction

Welcome to the Foreign Exchange Market!

- How does the exchange rate affect international trade?
- What determines exchange rates in the long run and short run?
- What role do interest rates play in exchange rate movements?



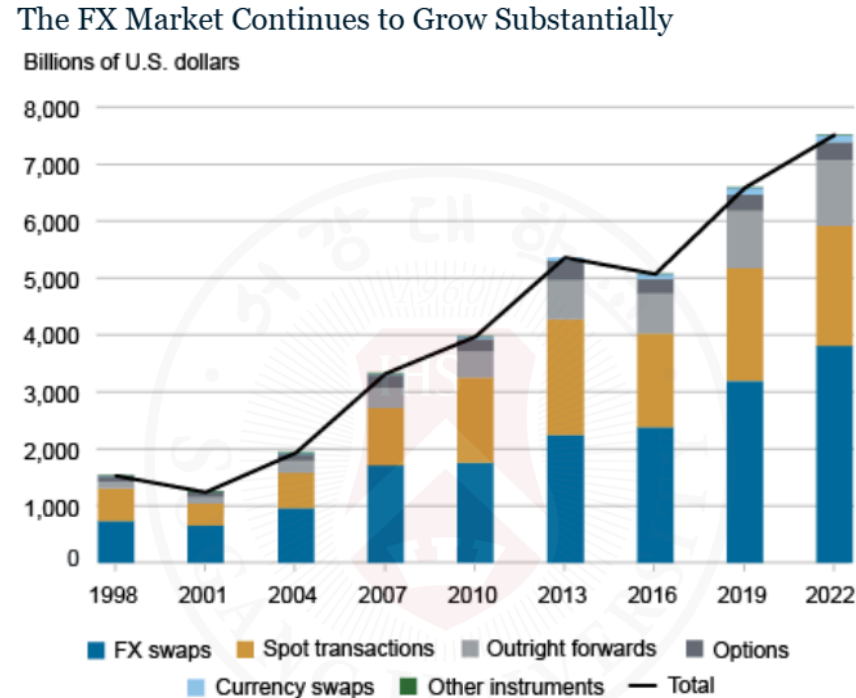
Definition and Importance

What is the Foreign Exchange Market?

- A market where different currencies are traded.
- Composed of banks, corporations, governments, and traders.
- Major transactions include:
 - Spot transactions: Immediate exchange of currencies.
 - Forward transactions: Agreement to exchange currencies at a future date.
- Facilitates international trade and investment by allowing for currency exchange.
- Influences import/export prices, financial flows, and investment decisions.
- Affects inflation, interest rates, and economic growth.
- FX markets operate 24 hours across global financial centers.

The FX market is the largest financial market globally, with turnover far exceeding global stock markets.

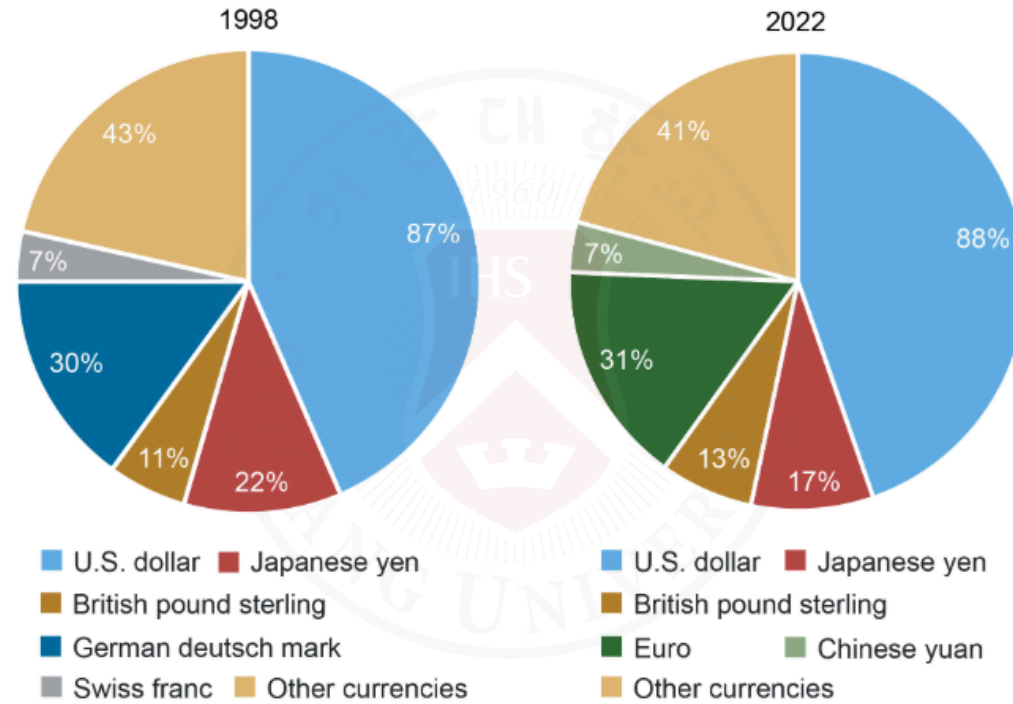
Size and composition of the FX Market



- Average daily turnover increased from \$1.5 to \$7.5 trillion between 1998 and 2022, with the increase occurring across both FX spot and FX derivatives. [link](#)

USD dominance in the FX Market

U.S. Dollars Remain the Dominant Currency in FX Transactions



Source: BIS Triennial Central Bank Survey, 1998 and 2022.

- The U.S. dollar (USD) is the most traded currency, accounting for 88% of all FX transactions. [link](#)

UK and US dominance in the FX Market

The U.S. and U.K. Continue to Dominate FX Trading

1998		2022	
United Kingdom	32.6	United Kingdom	38.1
United States	18.2	United States	19.4
Japan	7.0	Singapore	9.4
Singapore	6.9	Hong Kong	7.0
Germany	4.7	Japan	4.4

Source: BIS Triennial Central Bank Survey.

Note: Percent of total trading volume.

- The UK and US are the two largest FX trading centers, accounting for 38% and 19% of global FX turnover, respectively. [link](#)

Source: Liberty Street Economics

London dominates partly because of time-zone advantages between Asia and North America.

Participants in the FX Market

- Retail Traders
- Commercial Banks
- Central Banks
- Corporations
- Governments
- Investment Banks
- Hedge Funds

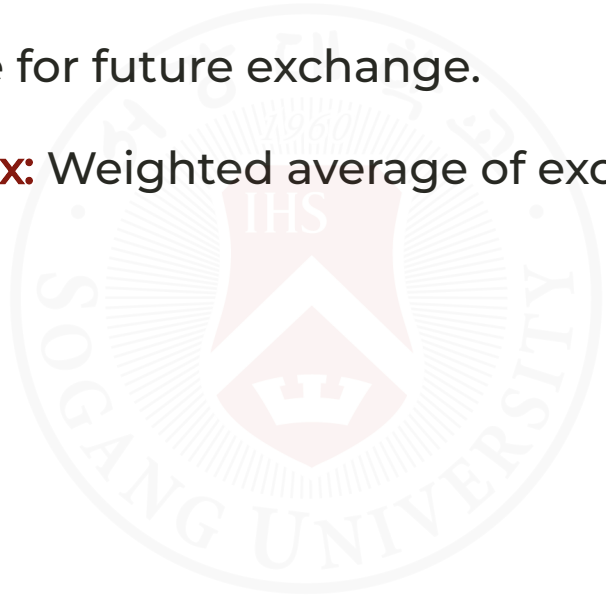
Who benefits the most and why?

High-frequency trading firms now account for a meaningful share of short-term FX trading volume.

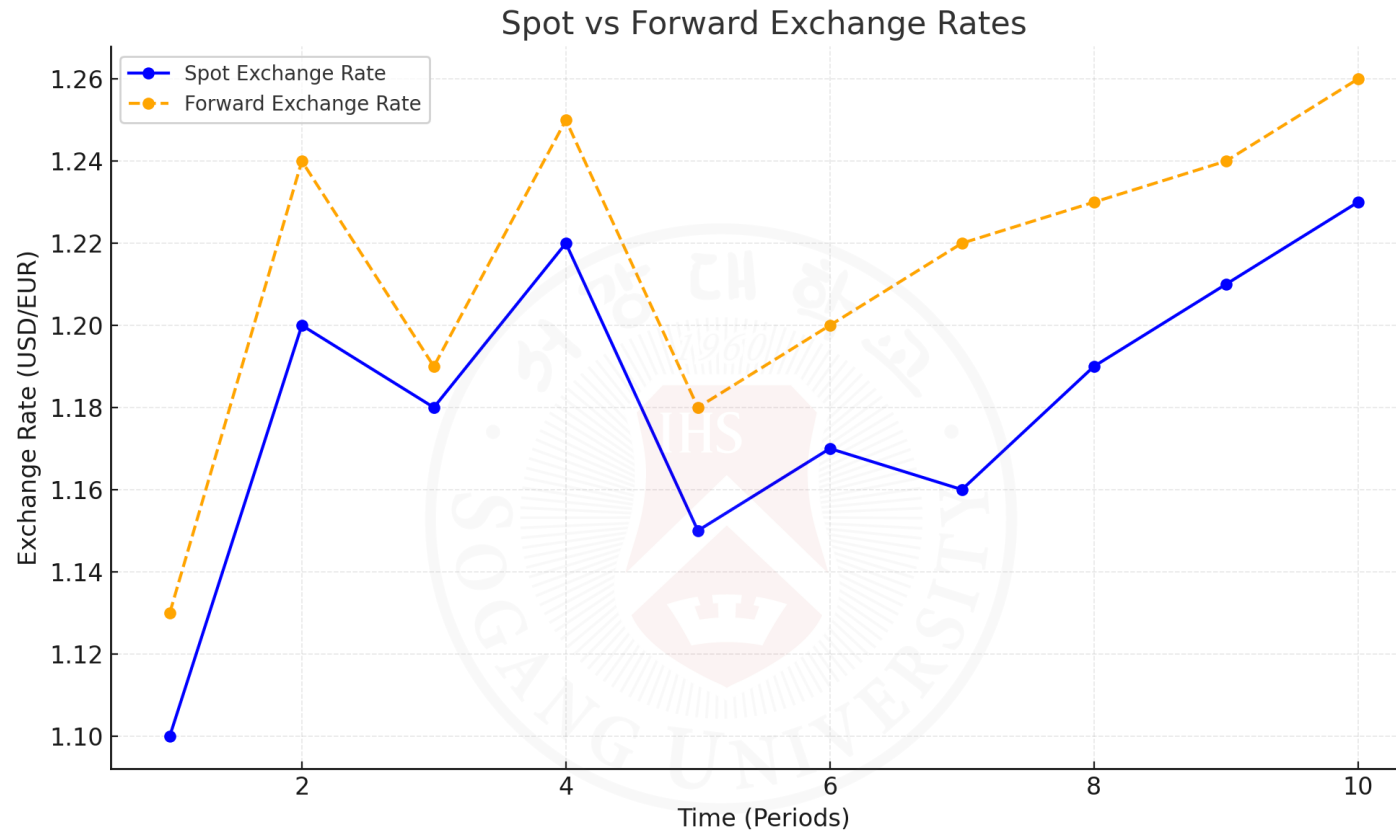


Key Terminologies

- **Exchange Rate (E):** The price of one currency in terms of another.
- **Spot Exchange Rate:** Rate for immediate exchange.
- **Forward Exchange Rate:** Rate for future exchange.
- **Effective Exchange Rate Index:** Weighted average of exchange rates with major trading partners.



Spot vs. Forward Exchange Rates



- Forward exchange rates are closely linked to interest rate differentials through covered interest parity.

Note: Covered interest parity states that the forward exchange rate should incorporate the interest rate differential between two countries to prevent

Types of FX Transactions

- **Spot Transactions:** Immediate exchange of currencies at the current rate.
- **Forward Transactions:** Agreement to exchange currencies at a future date.
- **FX Swaps:** involve exchanging currencies today and reversing the transaction later at a predetermined rate.
- **Options:** Contracts granting the right to buy/sell at a specific rate.

Why might a firm prefer a forward over a spot transaction?

Real vs. Nominal Exchange Rates

Nominal Exchange Rate (E): The price of one currency in terms of another.

Real Exchange Rate (Er): Adjusts the nominal rate for price level differences.

$$E_r = \frac{E \times P_{HC}}{P_{ROW}}$$

Where:

- E = Nominal Exchange Rate, foreign currency per unit of home currency
- P_{HC} = Price level in Home Country
- P_{ROW} = Price level in Rest of the World

Example: Real Exchange Rate Calculation

- HC: USA, ROW: Japan
- Exchange Rate (E): 150 yen/USD
- Price of U.S. wheat: 2 USD per bushel
- Price of Japanese wheat: 400 yen per bushel

$$E_r = \frac{150 \times 2}{400} = 0.75$$

Interpretation: 1 bushel of U.S. wheat costs 0.75 bushels of Japanese wheat.

Purchasing Power Parity (PPP)

- PPP states that identical goods should cost the same in different countries when expressed in a common currency.

$$E \times P = P_{ROW}$$

- PPP works better as a long-run tendency than as a short-run predictor.

Example: Big Mac Index

- If a Big Mac costs \$5 in the USA and 500 yen in Japan, the implied PPP exchange rate is $500 / 5 = 100$ yen/USD.
- Transportation costs, tariffs, taxes, and non-tradable goods prevent exact PPP.

Covered Interest Rate Parity (CIP)

Definition: The forward exchange rate fully incorporates the interest rate differential, leaving no arbitrage profit.

$$F = S \times \frac{1 + i_{HC}}{1 + i_{ROW}}$$

Or equivalently:

$$\frac{F - S}{S} \approx i_{HC} - i_{ROW}$$

Where F = forward rate, S = spot rate, i_{HC} = home interest rate, i_{ROW} = foreign interest rate.

- Uses the **forward exchange rate** — no exchange rate risk remains.
- A pure **arbitrage** relationship enforced by covered carry trades.
- Holds very closely in normal market conditions; broke down during 2008 GFC and COVID-19.

| CIP says: lock in the forward rate today → no risk, no excess return.

Uncovered Interest Rate Parity (UIP)

Definition: The expected change in the spot exchange rate equals the interest rate differential between two countries.

$$i_{HC} - i_{ROW} = \frac{E[S_{t+1}] - S_t}{S_t}$$

Rearranging for the expected future spot rate:

$$E[S_{t+1}] = S_t \times (1 + i_{HC} - i_{ROW})$$

Where $E[S_{t+1}]$ = expected future spot rate, S_t = current spot rate.

- Relies on **exchange rate expectations** — exchange rate risk is not hedged.
- If $i_{HC} > i_{ROW}$: the home currency is expected to **depreciate**.
- Empirically weak in the short run: the "**forward premium puzzle**" — high-interest currencies often appreciate rather than depreciate.

UIP says: higher rates today predict depreciation tomorrow — but markets frequently disagree.

CIP vs UIP: Key Comparison

Covered (CIP)

- Uses forward rate F
- No exchange rate risk
- Arbitrage enforced
- Holds tightly in practice

$$\frac{F - S}{S} \approx i_{HC} - i_{ROW}$$

Uncovered (UIP)

- Uses expected spot rate $E[S_{t+1}]$
- Exchange rate risk remains
- Expectation based
- Fails often short-run

$$\frac{E[S_{t+1}] - S_t}{S_t} \approx i_{HC} - i_{ROW}$$

Both conditions share the same right-hand side — the interest rate differential — but differ in whether exchange rate risk is hedged.

Wrap-Up and Discussion

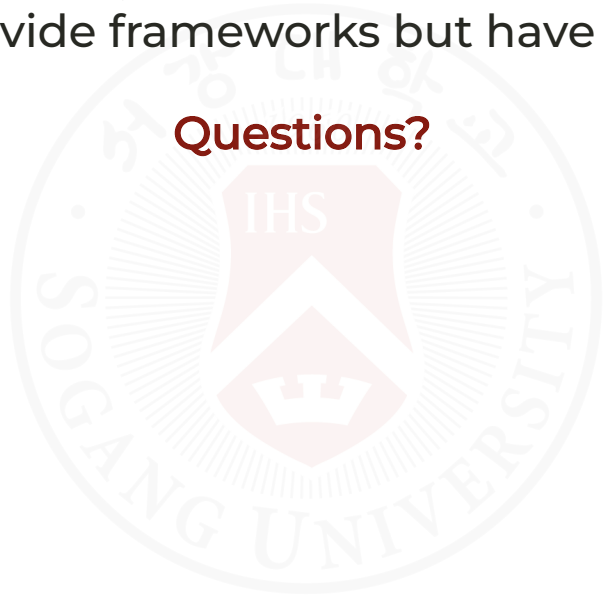
- What determines exchange rates in the short run and long run?
- How do interest rates and inflation impact exchange rates?
- How can businesses hedge against exchange rate risks?
- Exchange rates are strongly influenced by expectations and capital flows.



Closing and Q&A

- Key Takeaways:
 - FX markets are essential for global trade and investment.
 - Exchange rates are influenced by interest rates, inflation, and market expectations.
 - PPP and interest parity provide frameworks but have limitations.

Questions?

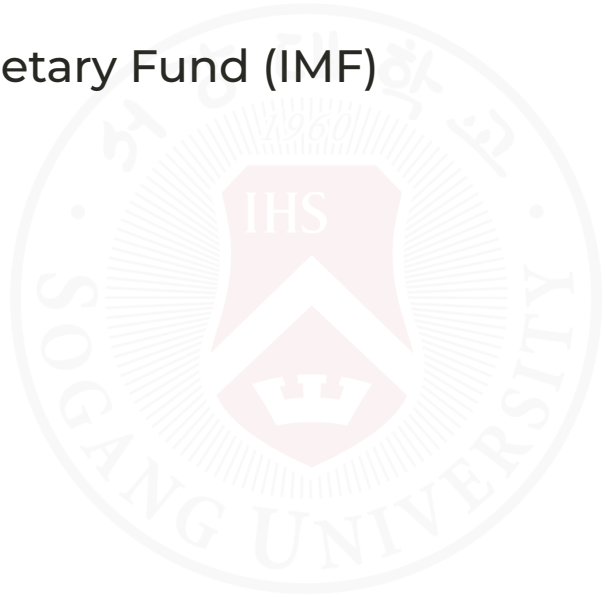


The seal of Gwangju University is a circular emblem. It features a central shield with a crown on top and the letters 'IHS' inside. The shield is set against a background of a sunburst or starburst pattern. The outer ring of the seal contains the university's name in Korean and English, along with the year '1960' at the top.

2. The International Financial System

Introduction to the International Financial System

- Overview of National Income and Product Accounting in an Open Economy
- Understanding the Balance of Payments
- Role of the International Monetary Fund (IMF)



National Income and Product Accounting for an Open Economy

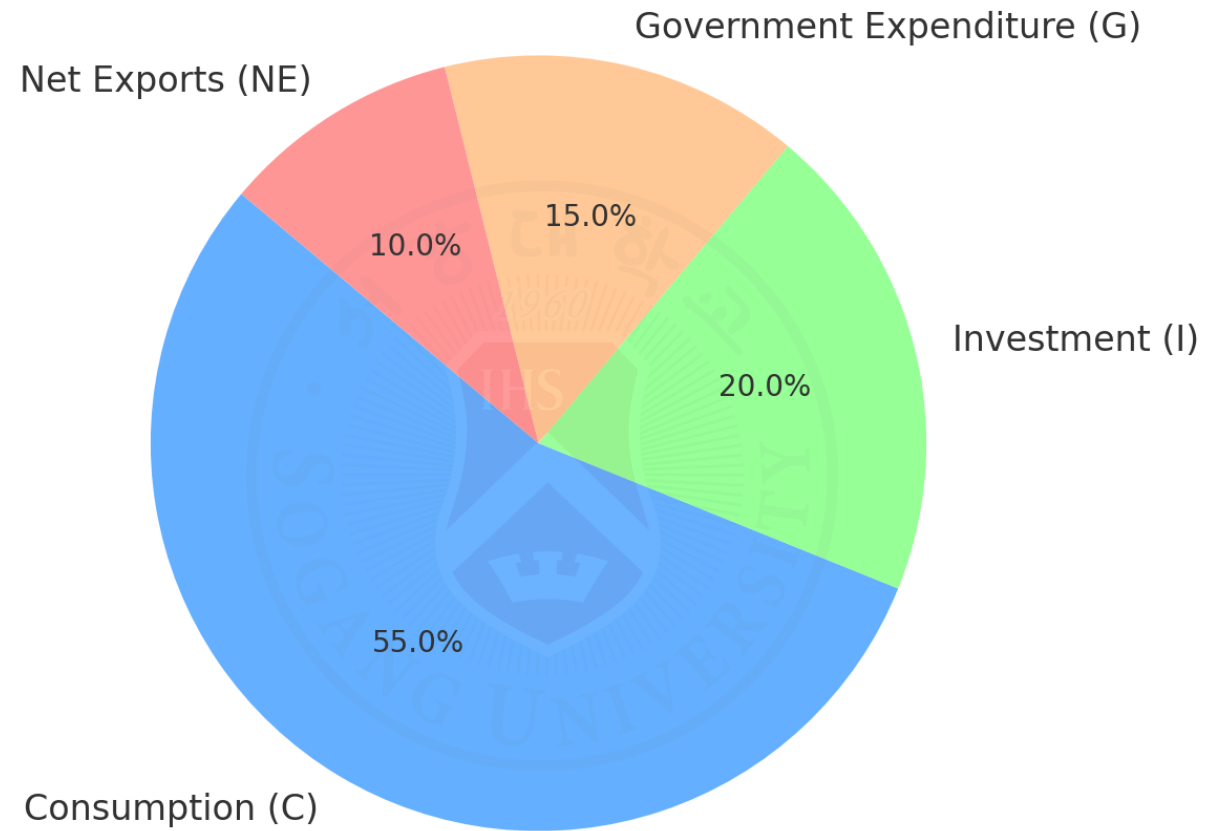
- Distinguish between GNP and GDP:
 - **GNP**: Total value of final goods/services produced using HC-owned factors of production.
 - **GDP**: Total value of final goods/services produced within the borders of the HC.
- Key Formula:

$$\text{GNP}(T) = \text{GDP}(T) + \text{NFP}(T)$$

- **NFP(T)**: Net factor payments from ROW to HC.

Modern international statistics often use GNI (Gross National Income) instead of GNP.

National Income Accounting Identity: $GDP = C + I + G + NE$



GNP vs. GDP



Value of all goods & services made by
a country's residents & businesses,
regardless of production location

Sound indicator of financial well-being
of Americans & U.S.-based
multi-national corporations



Measures production inside
of a country, no matter
who makes it

Sound indicator of health of
U.S. economy

 the balance

Link Source: the balance

National Income Accounting Identity

- National Income Accounting Identity:

$$\text{GDP}(T) = C(T) + I(T) + G(T) + \text{NE}(T)$$

- Key Terms:
 - **C(T)**: Consumption Expenditure
 - **I(T)**: Investment Expenditure
 - **G(T)**: Government Expenditure
 - **NE(T)**: Net Exports (EX - IM)



Example of GNP and GDP Calculation

- Suppose a country produces:
 - \$100 million worth of goods/services domestically (GDP)
 - \$20 million received as income from abroad
 - \$10 million paid to foreign factors
- GNP Calculation:

$$\text{GNP} = 100 + (20 - 10) = 110 \text{ million}$$

Balance of Payments: Definition and Structure

- BOP Accounts:
 1. Current Account (CA): Net exports, factor income, transfers
 2. Capital and Financial Account (KFA): Asset transfers and financial transactions
 3. Official Reserve Transactions
- Simplified teaching identity:

$$CA + KFA \approx 0$$

where **KFA** groups capital, financial, and reserve-related balancing flows.

Balance of Payments: Accounting Identity (Advanced)

In full accounting form:

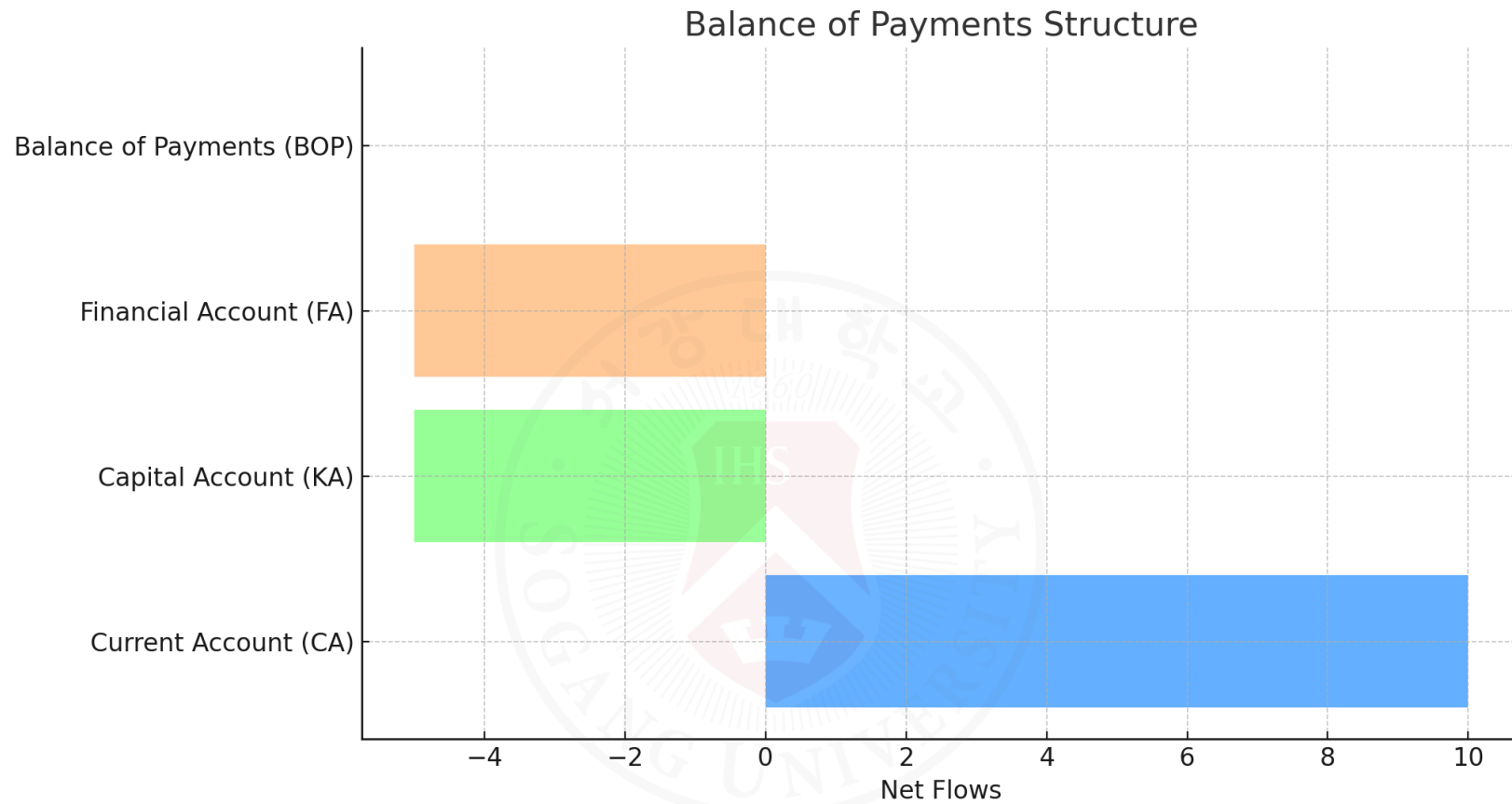
$$CA + KA + FA + ORT + EO = 0$$

Where:

- *CA*: current account balance
- *KA*: capital account balance
- *FA*: financial account balance
- *ORT*: official reserve transactions
- *EO*: errors and omissions (statistical discrepancy)

Interpretation:

- The balance of payments must sum to zero once all recorded and unrecorded transactions are included.
- A deficit in one component must be offset by a surplus in other components and/or reserve changes.



Role of the International Monetary Fund (IMF)

- Established in 1944 as part of the Bretton Woods Agreement
- Key Functions:
 - Promote international monetary cooperation
 - Provide financial stability
 - Offer financial assistance to member countries
 - Monitor economic policies and provide technical assistance

What does the IMF do?

The IMF fosters international financial stability by offering:



POLICY ADVICE

Monitoring economic and financial developments and advising countries.

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FINANCIAL ASSISTANCE

Loans and other financial aid to member countries.

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CAPACITY DEVELOPMENT

Technical assistance and training to help governments to implement sound economic policies.

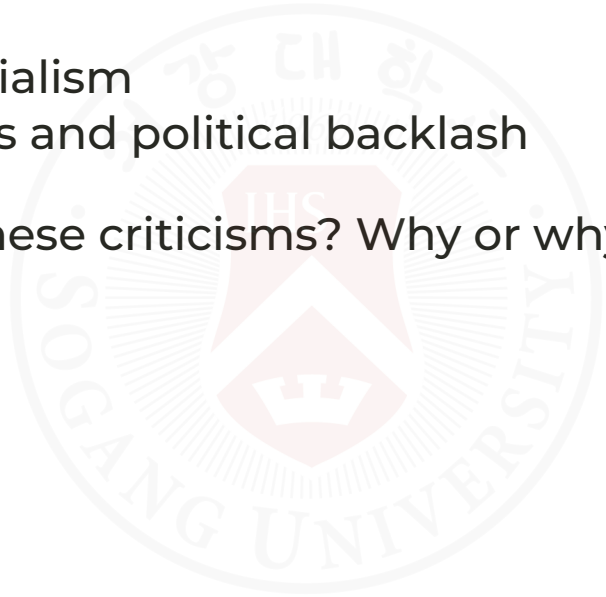
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Link

| The IMF increasingly focuses on financial stability, climate risk, and debt sustainability.

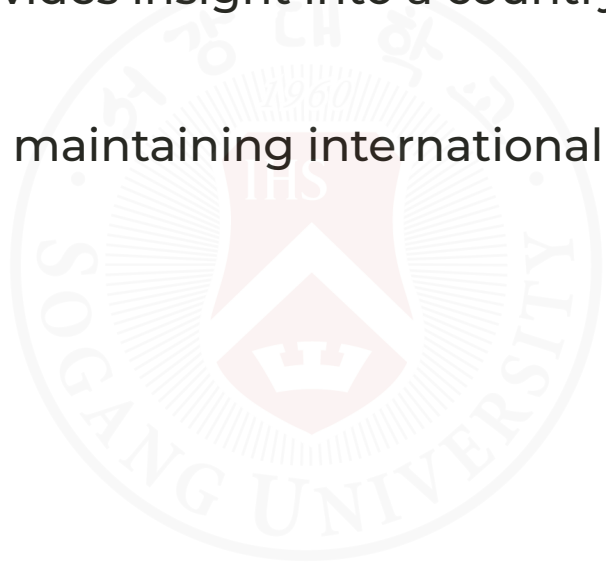
IMF Controversies and Criticism

- Criticisms of IMF Interventions:
 - Criticism that IMF austerity measures may deepen recessions
 - Moral hazard concerns
 - Perceived economic imperialism
 - Concerns about social costs and political backlash
- Discuss: Do you agree with these criticisms? Why or why not?



Conclusion

- National Income and Product Accounting is essential for understanding a country's economic performance in a global context.
- The Balance of Payments provides insight into a country's financial position relative to the world.
- The IMF plays a critical role in maintaining international financial stability.



3. In-Class Activity & Discussion



The background of the slide features a large, faint, circular seal of Gangneung University. The seal contains the university's name in Korean and English, the year 1960, and a central shield with the letters 'IHS' and a crown.

Course Roadmap

Looking Ahead: Course Roadmap

We are entering the final phase of the course. Here's what's coming:

Session Schedule & Topics

- **(May 21) ESG and Sustainable Finance**
 - The role of financial institutions in promoting sustainability
 - Reading will be assigned
- **(May 28) FinTech Revolution**
 - Guest Speaker (in-person)
 - How technology is reshaping financial services
- **(June 4) AI & Machine Learning in Finance**
 - Guest Speaker (online)
 - Final Exam Review Session
- **(June 11) Student Project Presentations**
 - Showcase your applied research and insights
- **(June 18) Final Examination**
 - Comprehensive assessment of course material

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Any Questions?

Thank you for your attention and participation!